

stages. This can then be aligned with the storage, server and the application infrastructure in the organisation to build the critical parameters governing the definition of the various tiers.

The identification of different categories of data and assigning them to the appropriate types of storage, play an important part in the storage infrastructure tier-ing, which reduces cost as well as optimises the performance levels required by the applications. Functional categories include the type of back-ups, retrieval, type of protection needed, availability requirements, and so on.

Non-functional attributes include performance requirements, frequency of usage of the data stored, and so on. The tier-ing process is ongoing and there are commercially available Storage Resource Management (SRM) applications that can be leveraged to achieve this. In the tier-ing activity, as illustrated in Figure 2, the different vendor storage platforms can be integrated to form a homogeneous pool of storage, using technologies such as virtualisation on top of which a tiered storage model can be implemented based on the criticality of data stored, performance requirements, etc.

For example, Tier 1 comprises business-critical data that has a high frequency of access and needs to be on premium storage that can address the high performance requirements of this type of data. Tier 2 data, which is relatively lower priority, may be stored on media that

guarantee availability, but at a lower Quality of Service in conventional Storage Area Networks (SANs). Tier 3 is low priority data that is not frequently accessed [such as e-mail archives]. This type of data can be stored on cheaper storage where the performance requirements are less.

Key attributes to be considered when building a tier-ing model from a server perspective are the number of processors, availability of memory, number of I/O ports, etc. In a similar manner, application tier-ing can be achieved by separating business applications into multiple tiers based on functional parameters such as application binaries, criticality of data, etc, and non-functional parameters such as performance and the Quality of Service (QOS) needed.

Infrastructure tier-ing is a holistic capacity planning model that aims at assessing the current and future capacity requirements of a business' IT infrastructure and determines a comprehensive tiered model to handle business applications in the most effective way. It is very important to be aware of all the building blocks of the infrastructure when designing a tiered storage model, without which the true benefits cannot be enjoyed. 

By: Venkat Thummisi

The author is a senior manager, Technology, at Cognizant.



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